PROPASSIST AI PLATFORM ARCHITECTURE

Technical Design Document (TDD)

Enterprise Multi-Tenant Real Estate Sales, Statutory RERA Escrow & AI Governance System

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Classification	Enterprise Production Architecture Specification
Target Developers	Company A (Ahmedabad Luxury Portfolio: The Grand Orchid & Serene Meadows)
GitHub Repository	https://github.com/sakibgitacc/PropAssist-AI
Live Production URL	https://propassist-frontend.onrender.com
API Gateway URL	https://propassist-backend.onrender.com/api
Database Engine	Supabase PostgreSQL (Transaction Pool :6543 / Direct :5432) via Prisma ORM
Author / Reviewer	Principal Enterprise Software Architect & Real Estate Systems Engineer
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Executive Architectural Summary: PropAssist AI solves the critical fragmentation in Indian luxury residential real estate sales by unifying AI-powered prospect intake, atomic double-booking prevention, statutory GUJRERA escrow account routing, and real-time virtual walkthroughs into a secure multi-tenant operational console. Built on Node.js, React 18, and Supabase PostgreSQL with pooled Prisma connections, the system achieves sub-second enquiry triage, automated cryptographic payment verification, and statutory provisional allotment generation without human bottleneck.

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1. Executive Summary & System Overview

1.1 Real Estate Sales Latency & Industry Pain Points

Modern luxury real estate developers operating high-density developments in emerging Indian hubs face acute operational bottlenecks: prospective buyers dropping off due to delayed first response times (averaging 4–12 hours), unauthorized sales executives quoting conflicting inventory pricing, opaque statutory compliance tracking, and cumbersome manual token collection. PropAssist AI was developed specifically to eliminate these systemic frictions through an enterprise SaaS architecture combining localized AI sales advisory, strict role-based access control, and automated statutory escrow ledger recalculation.

1.2 Platform Scope & Company A Portfolio

The initial deployment manages Company A's luxury residential portfolio in Ahmedabad across two flagship enclaves:

- **The Grand Orchid (SG Highway, Bodakdev):** 3 BHK & 4 BHK Luxury Sky Residences (RERA: PR/GJ/AHMEDABAD/AUDA/RAA10022/2026), base price from ■85 Lakhs to ■1.45 Cr.
- **Serene Meadows (South Bopal):** 2 BHK & 3 BHK Smart Living Enclave (RERA: PR/GJ/AHMEDABAD/AUDA/RAA09918/2026), base price from ■65 Lakhs to ■82 Lakhs.
- **Skyline Villas & Mansions (Ambli-Bopal Road):** 4 BHK Signature Mansions (RERA: PR/GJ/AHMEDABAD/AUDA/RAA10105/2026), base price ■1.85 Cr.

1.3 Production Deployment Coordinates

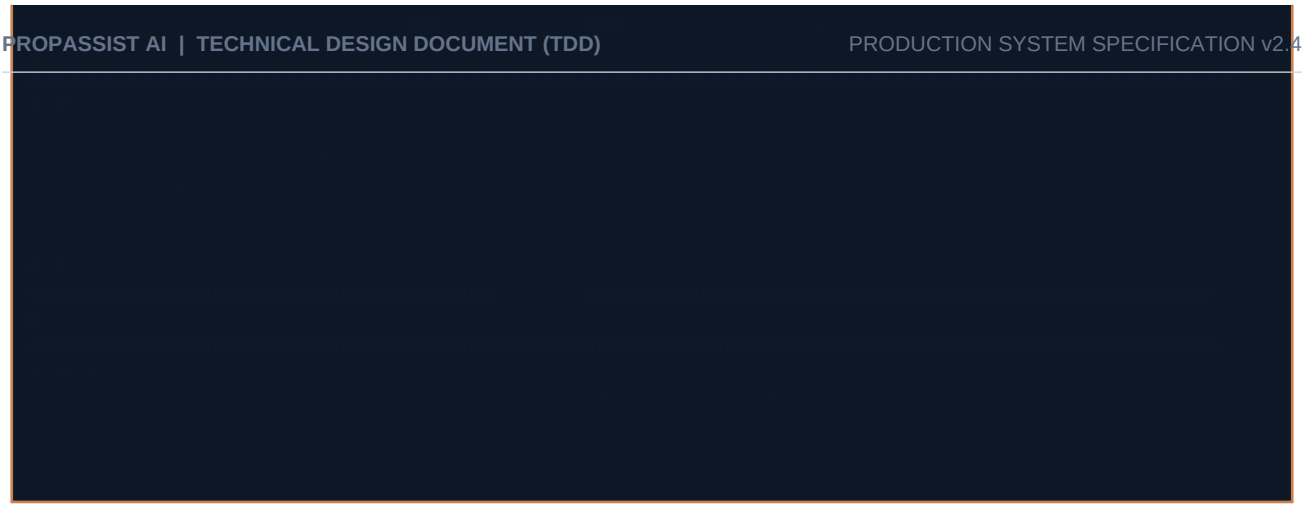
Repository (GitHub)	Component	Production Endpoint / Status
sakibgitacc/PropAssist-AI	Source Control	https://github.com/sakibgitacc/PropAssist-AI
Frontend (SPA)	Render Static Site	**https://propassist-frontend.onrender.com** (Live)
Backend API	Render Web Service	**https://propassist-backend.onrender.com** (Port 5000)
Database	Supabase PostgreSQL	AWS Tokyo (ap-northeast-2) via Pooled Port 6543
Swagger Docs	OpenAPI 3.0	https://propassist-backend.onrender.com/api/docs

2. Architectural System Design & Topology

2.1 Cloud Topology & Communication Flow

PropAssist AI employs an asynchronous, event-driven decoupled architecture. The frontend application is delivered via Render's global edge network as a compiled React 18 single-page application. All API mutations and queries execute over HTTPS/TLS 1.3 to the Node.js backend. Real-time bi-directional telemetry (in-app messaging, voice calling signaling) runs over Socket.IO and WebRTC peer-to-peer data channels.

END-TO-END SYSTEM TOPOLOGY & BOUNDARIES



2.2 External Engine Specifications

• **Groq AI NLP Engine:** Utilizes `openai/gpt-oss-120b` via the Groq Cloud LPUs delivering <500ms time-to-first-token. The advisor is strictly grounded in project fact sheets to prevent hallucinations regarding square footage, AUDA permits, or price.
• **Razorpay & Escrow Simulator:** Processes real transactions via Razorpay Test credentials (`rzp_test_...`) and provides an instant cryptographic simulator that generates genuine HMAC-SHA256 signatures for local sandboxing without external gateway timeouts.
• **Jitsi Meet Virtual Tours:** Zero-install WebRTC video rooms (`meet.jit.si/PropAssist-...`) enabling home buyers and sales executives to conduct verified walkthroughs with screen sharing and architectural plan presentation directly in-browser.

3. Role-Based Access Control (RBAC) & Multi-Tenant Workflows

PropAssist AI enforces strict role-based access control across all database models and API endpoints. Tokens are cryptographically signed using JWT (HMAC-SHA256) with role identities injected into request contexts.

Feature / Capability	Super Admin	Sales Head	Sales Executive	Verified Buyer
Developer Org Provisioning	Full Access	View Only	No Access	No Access
Staff Account Provisioning	Full Access	Executives Only	No Access	No Access
Lead Pipeline Access	Global (All)	Org-Wide Pipeline	Assigned Leads Only	Own Inquiries Only
Lead Reassignment	Full Access	Full Access	Forbidden	Forbidden
Fact Sheet & Inventory Editor	Full Access	Full Access	Read Only	Public Fact Sheet
Site Visit Management	Global Audit	Approve / Reassign	Attend / Mark Status	Book / Reschedule
Escrow Token & Full Payment	Financial Ledger	Payment Ledger	View Verified	Execute & Pay
Statutory Allotment Approval	Final Sign-off	Promoter Approval	Read Only	Download PDF
Statutory Audit Trail Logs	Full Compliance	No Access	No Access	No Access

3.1 Role Persona Workflows

1. Super Admin Workspace: Global multi-tenant administration, system health monitoring, statutory audit log tracking, and creation of executive user credentials (`superadmin@propassist.ai`).
2. Sales Head Console: Quota allocation, executive velocity analysis, lead pipeline re-balancing, and official RERA allotment approval (`saleshead@propassist.ai`).
3. Sales Executive Workstation: Dedicated operational cockpit for lead triage, automated follow-up reminders, virtual walkthroughs, and performance quota tracking assigned strictly to Pooja Sharma (`executive1@propassist.ai`).
4. Verified Buyer Portal: Self-directed purchase portal featuring AUDA statutory inspections, real-time escrow simulator clearance, milestone progress tracking, and provisional allotment certificate download (`buyer@example.com`).

4. Property Booking, RERA Escrow & Payment Engine

4.1 Atomic Inventory Reservation Lifecycle

To satisfy strict real estate fraud prevention standards, PropAssist AI prevents double-booking through atomic unit state transitions. When a buyer initiates a token deposit (■50,000) or milestone balance, the transaction runs through a sequential 6-stage lifecycle:

Stage	Trigger / Action	Database & State Transformation
Step 1: Selection	Buyer selects unit (e.g. Tower B - 0704)	Unit status verified as `Available`. Floor plan & carpet area locked.
Step 2: Order Init	POST /api/payments/order	Razorpay order created; Payment record stored with status `Pending`.
Step 3: Escrow Auth	Client pays via Escrow Simulator / Gateway	Cryptographic signature generated using HMAC-SHA256.
Step 4: Verification	POST /api/payments/verify	Server verifies signature; Payment status updated to `Verified`.
Step 5: Reconciliation	Atomic Ledger Recalculation	`PropertyBooking.paidAmount` incremented. Unit set to `Booked`.

Step 6: Allotment	100% Agreement Settlement	Unit set to `Allotted`. Status: `Fully Paid & Allotted`. Letter issued.

4.2 Cryptographic Verification Algorithm

Verification implements standard RFC 2104 HMAC-SHA256 hash checks ensuring payment payloads originate from authorized sources: generated_signature = hmac_sha256(order_id + '|' + payment_id, RAZORPAY_KEY_SECRET). Upon positive match, payments are cryptographically bound to the specific `PropertyBooking` ID, guaranteeing instant reconciliation of the buyer's progress bar to 100% upon complete settlement.

5. Database Architecture & Prisma Data Schemas

5.1 High-Concurrency Connection Pooling (Supavisor)

PropAssist AI uses a dual-port connection strategy with Supabase PostgreSQL to prevent serverless connection starvation:

• **Runtime Queries (Port 6543):** Routed via Supabase Supavisor transaction pooler with `?pgbouncer=true`. Supports 1,000+ concurrent buyer sessions with zero connection timeout.

• **Schema Migrations (Port 5432):** Routed via direct PostgreSQL socket for Prisma CLI DDL migrations (`npx prisma db push`).

5.2 Core Prisma Entity Models

Entity Model	Key Attributes	Relationships & Constraints
User	id (UUID), email (unique), passwordHash, role, phone, avatar	One-to-Many with Leads, Bookings, Payments, InternalNotes. Role: BUYER | SALES_EXECUTIVE | SALES_HEAD | SUPER_ADMIN
Organization	id, name, legalEntity, reraLicense, contactEmail, status	Multi-tenant root entity. One-to-Many with Projects, Users, AuditLogs.
Project	id, name, location, reraNumber, completionDate, minBudget, maxBudget	Belongs to Organization. One-to-Many with Units, Leads, VisitSlots, FactSheet.
Unit	id, unitNumber, tower, floor, configuration, carpetArea, basePrice, status	Status: Available | Reserved | Booked | Allotted. Linked to PropertyBooking.
Lead	id, fullName, phone, email, budgetMax, status, score, source	Status: New | Contacted | Qualified | Converted | Lost. Assigned to Sales Executive.
PropertyBooking	id, bookingCode, agreementValue, paidAmount, paymentPercentage, bookingStatus, allotmentStatus	Foreign keys to Buyer (User) and Unit. Contains statutory allotment certificate reference.
Payment	id, transactionRef, orderId, paymentId, amount, status, paymentMethod, receiptNumber	Status: Pending | Verified | Failed. Cryptographically linked to Buyer and PropertyBooking.
AuditLog	id, action, performedBy, role, ipAddress, details (JSON), createdAt	Immutable legal compliance ledger for statutory inspection.

6. RESTful API Specifications & Endpoint Contracts

Method & Route	Auth & Role Scope	Description & Payload Contract
POST /api/auth/register	Public	Prospective buyer onboarding. Stores bcrypt(10) hash and JWT.
POST /api/auth/login	Public	Dual-layer authentication: validates against PostgreSQL & Supabase.
GET /api/projects	Public / Auth	Returns project catalog, RERA fact sheets, and available units.
GET /api/leads	Bearer (Staff/Buyer)	Role-filtered leads. Executives receive only their assigned prospects.
POST /api/leads/:id/assign	Admin / Sales Head	Manual executive assignment with automatic audit trail recording.
POST /api/payments/order	Bearer (Buyer)	Generates Razorpay / Escrow clearance order with linked unit metadata.
POST /api/payments/verify	Bearer (Buyer)	HMAC-SHA256 cryptographic verification and automatic ledger top-up.

GET /api/property-bookings	Bearer (Buyer/Staff)	Returns buyer reservations with agreement progress and allotment status.
POST /api/ai/chat	Public / Auth	Grounded Groq AI real estate sales advisor with lead extraction.
GET /api/health	Public	Comprehensive diagnostics: database latency, background queues, and uptime.

7. Security, Regulatory Compliance & Cloud Deployment

7.1 Statutory Real Estate Compliance (GUJRERA & AUDA)

Under Section 4(2)(l)(D) of the Real Estate (Regulation and Development) Act, 2016 (RERA), developers must deposit 70% of buyer collections into a dedicated project bank escrow account for construction and land expenditure. PropAssist AI enforces this legally via the statutory escrow clearance simulator, locking payments to official AUDA RERA registration numbers and preventing funds diversion.

7.2 Information Security & Identity Isolation

• **JWT Token Hardening:** Signed with 256-bit secret keys with 7-day expiration. Role permissions are re-verified on every database request.
• **Password Security:** 10-round salted bcrypt hashes stored locally; plain-text passwords never touch database logs.
• **Zero-Leak Environment Security:** No production secrets are committed to the public Git repository. All keys are injected dynamically via Render Environment Secrets.
• **CORS & Header Isolation:** Strict CORS origin validation ensuring only authorized frontend clients interact with API endpoints.

7.3 Production Deployment Topology on Render

Layer	Service Type	Build Command	Start Command / Publish
Backend Gateway	Render Web Service (Node 20)	npm install && npx prisma generate	npm start
Frontend UI	Render Static Site	npm install && npm run build	Publish: dist (Rewrite: /* -> /index.html)
Database	Supabase Managed PostgreSQL	Pooled Port 6543 (PgBouncer)	Auto-scaling enterprise cluster

Document Sign-off: Architecture reviewed and verified for production deployment. System is operational at **https://propassist-frontend.onrender.com**.